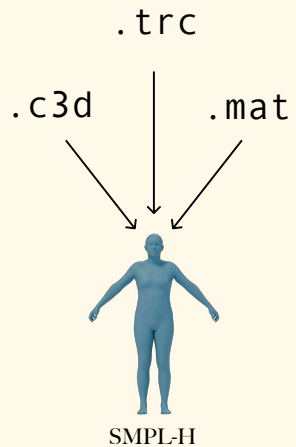
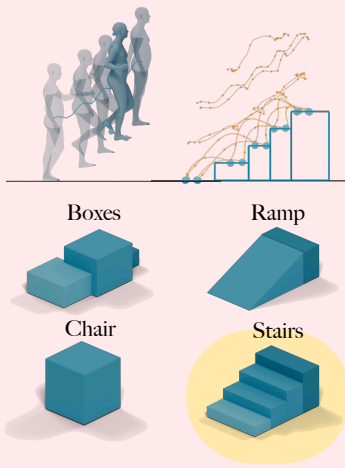


TERRA

Motion Inputs



Terrain Reconstruction



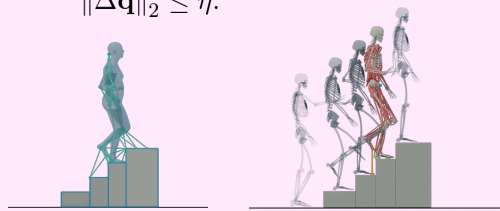
Biomechanics-aware Motion Retargeting

$$\arg \min_{\Delta \mathbf{q}} \underbrace{E_{\text{mesh}}}_{\text{interaction}} + \underbrace{E_{\text{temp}} + E_{\text{pose}}}_{\text{regularization}} + \underbrace{\sum_{m \in \mathcal{M}} E_m}_{\text{biomechanics}}$$

$$\text{s.t. } \bar{\phi} + J_{\phi} \Delta \mathbf{q} \geq -\epsilon,$$

$$\mathbf{q}^{\text{lb}} \leq \mathbf{q} + \Delta \mathbf{q} \leq \mathbf{q}^{\text{ub}},$$

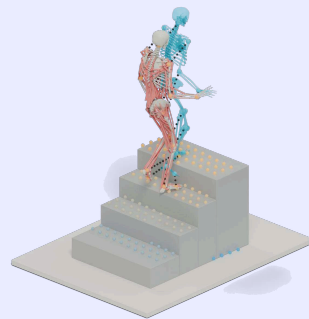
$$\|\Delta \mathbf{q}\|_2 \leq \eta.$$



RL Training

$$\mathbf{a}_t^{\text{muscle}} = \pi_{\theta}(\mathbf{o}_t^{\text{prop}}, \mathbf{h}_t, \mathbf{g}_t)$$

$$\theta \leftarrow \theta + \eta \nabla_{\theta} \mathcal{L}_{\text{PPO}}(\theta)$$



Biomechanical Validation

